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ARTIFICIAL INTELLIGENCE (SECJ3553)

SECTION 03

GROUP 2

**PROJECT:
ECOROUTE**

**THEME:
SMART CITY**

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Problem Statement

The current waste management framework in Malaysian cities still follows a rigid schedule for example collections on every Monday and Thursday without considering how much rubbish is being produced at any given time. This inefficiency forces the City Councils often end up spending unnecessary fuel and budget to collect bins that are barely filled, while other high-density areas experience overflowing bins during peak periods. This not only wastes resources but also leads to unpleasant and unhygienic conditions that can affect public health and the well-being of residents.

AI Solution

EcoRoute is an AI-powered system designed to transform traditional waste collection into a fully intelligent, demand-driven process. Instead of following fixed schedules, EcoRoute uses real-time sensor data and predictive analytics to determine when each rubbish bin actually needs servicing. By continuously monitoring fill-levels, the system identifies bins that are almost full and prioritizes them for collection.

A route optimisation engine then generates the most efficient collection paths for waste trucks, reducing unnecessary trips to empty bins and preventing overflowing bins in high-usage areas. The AI model also learns from historical waste patterns, such as daily trends, seasonal surges, or location-specific usage to predict when individual bins are likely to reach 80% capacity, enabling proactive planning rather than reactive cleanup.

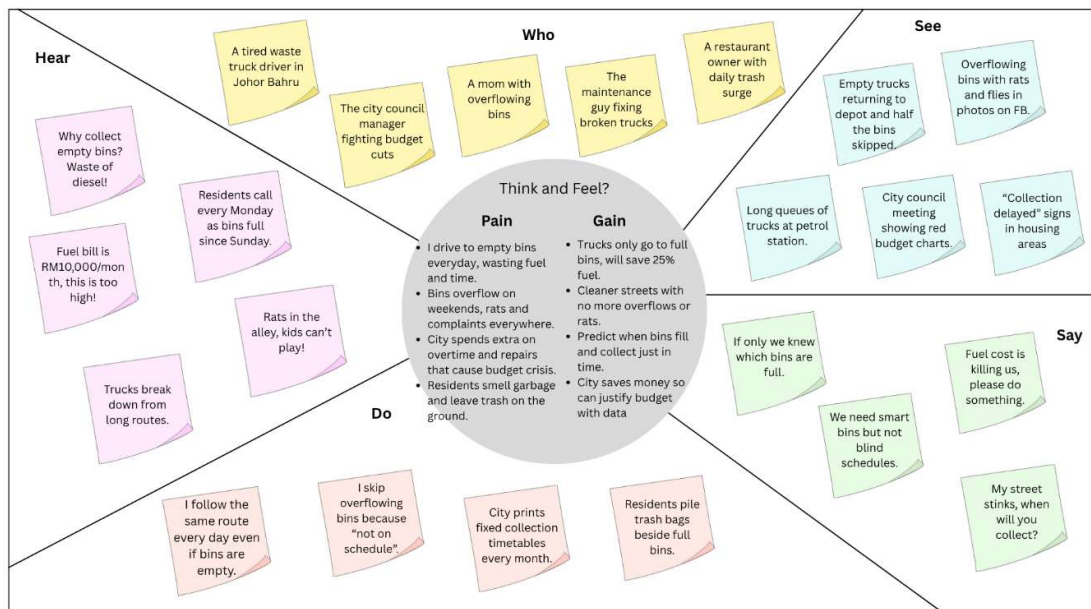
Ultimately, EcoRoute helps local councils and city management teams reduce operational and fuel costs, supports waste collection crews with clearer and more practical routes, and ensures cleaner, more hygienic neighbourhoods for residents. The solution contributes directly to sustainability goals by decreasing carbon emissions and minimizing wasted resources, making waste management smarter, greener, and more efficient.

Goal of AI Solution

The main goal of *EcoRoute* is to enhance the efficiency and sustainability of city waste management in Malaysian cities by using AI to monitor and predict bin fill-levels in real time. Through this system, the project aims to:

1. Support waste management crews and city administrators in identifying which bins need collection, reducing unnecessary trips.
2. Improve operational efficiency by generating AI-optimized collection routes based on real-time and estimated bin occupancy.
3. Minimise fuel consumption and carbon emissions, contributing to a greener and more sustainable urban environment.
4. Empower city authorities with data-driven insights to optimize waste collection planning, resource allocation, and overall service efficiency.

Process of Emphasis in Design Thinking



We empathized with waste crews, city councils and residents by stating their daily struggles in the sticky note sketch. The 'Pain and Gain' circle shows pains like wasted fuel and overflowing bins, with gains like smart routes and cleaner streets. 'Who' includes drivers and residents, 'Hears' captures complaints, 'Does' notes fixed schedules, 'Sees' spots empty trucks, and 'Says' reflects frustration like 'Why collect empty bins?' This aims for a predictive, on-demand collection system.

Process of Define in Design Thinking

User	Need	Insight
City Council	Need 1: To optimize the waste collection budget by reducing fleet operational costs like fuel, vehicle maintenance, and staff overtime Need 2: Needs verifiable data on collection efficiency to report to the city council and justify her department's budget.	Insight 1: The current fixed-route system is fundamentally inefficient because it treats all bins as equal. This fix scheduling wastes significant resources by forcing crews to drive to stop at, and service bins that are only 20% full, while simultaneously failing to cope with sudden surges in other areas.
A resident living in a high-density apartment complex	Need 1: To have reliable and sanitary access to waste disposal bins that are not overflowing. A resident needs to be able to dispose of his trash without experiencing nasty smells, rats or having to leave his trash bags on the ground, which he feels is a public health hazard.	Insight 1: The problem is not only that the bins get full, but that they get predictably. The existing static schedule such as collection every Monday or Thursday totally ignores the residence's actual usage habits such as higher waste generation on weekends. The bins are likely to overflow due to the inflexible schedules.
Waste Truck Driver	Need 1: To have shorter, smarter daily routes that reduce physical fatigue and driving time. Need 2: To avoid unnecessary maintenance on trucks from excessive, low-value driving, which leads to breakdowns.	Insight 1: The inefficiency of the fixed-route system has a direct human cost. Drivers are wasting an average of 2 hours per day driving to and stopping at empty bins, leading to burnout and fatigue. Insights 2: Financial effect occurs. On days when high-density bins overflow due to the bad schedule, workers

		are forced into overtime to clean up the rubbish, which directly increase the City Council's operational costs.
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